

MUZHI, Russian Federation

WMO: 234260

Lat: 65.404N

Lon: 64.703E

Elev: 62

StdP: 14.66

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.3	-37.4	-46.2	0.4	-41.7	-41.5	0.5	-37.1	24.5	9.4	21.1	11.4	2.2	10	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	79.1	66.7	74.8	64.4	70.9	62.0	68.6	76.0	65.9	72.7	63.3	69.5	7.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	95.5	73.2	63.2	87.0	69.8	60.4	78.5	67.5	32.9	75.7	30.8	72.9	28.8	69.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	20.8	18.2	DB	-46.5	85.2	5.6	3.2	-50.5	87.5	-53.8	89.4	-56.9	91.2	-61.0	93.6	
			WB	-46.1	72.7	5.0	3.2	-49.7	74.9	-52.6	76.8	-55.5	78.6	-59.1	80.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.4	-7.2	-6.1	10.1	23.2	36.2	52.3	60.2	53.7	43.5	29.5	8.5	-1.9
	DBStd	26.43	16.49	16.89	14.24	12.14	10.17	8.91	7.70	6.47	6.95	9.61	15.69	17.33
	HDD50	9651	1774	1571	1237	804	445	77	31	216	635	1244	1610	2075
	HDD65	14525	2239	1991	1702	1253	892	390	191	354	645	1099	1694	2075
	CDD50	660	0	0	0	1	18	147	325	146	22	1	0	0
	CDD65	59	0	0	0	0	0	10	44	5	0	0	0	0
	CDH74	431	0	0	0	0	3	82	309	36	1	0	0	0
	CDH80	78	0	0	0	0	0	12	61	5	0	0	0	0
(14) Wind	WSAvg	7.2	5.8	6.1	7.3	8.3	8.8	8.7	7.8	7.4	7.1	7.2	6.4	5.8
(15) Precipitation	PrecAvg	21.5	1.1	0.9	1.6	1.5	1.8	2.3	3.0	2.7	1.9	2.2	1.4	1.1
	PrecMax	26.7	2.0	2.0	10.4	3.2	4.3	5.1	6.9	7.0	5.0	3.4	2.8	2.2
	PrecMin	17.6	0.3	0.1	0.2	0.4	0.6	0.4	1.2	0.8	0.6	0.3	0.2	0.4
	PrecStd	2.5	0.5	0.5	1.8	0.7	0.9	1.2	1.3	1.6	0.9	0.7	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	32.2	41.3	56.0	72.9	81.2	85.7	78.6	69.7	54.6	38.4	33.0
		MCWB	30.1	29.6	35.2	46.6	57.3	65.3	69.7	65.4	60.1	49.1	35.2	30.2
	2%	DB	24.9	27.3	36.9	46.3	64.7	75.9	81.4	72.0	62.2	49.0	34.6	29.1
		MCWB	22.9	25.4	32.3	39.4	53.2	62.7	68.2	62.9	55.6	44.9	32.9	27.3
	5%	DB	20.6	22.9	34.0	42.3	57.8	71.1	77.4	68.3	57.4	44.9	32.1	25.2
		MCWB	19.2	21.5	30.5	36.6	48.9	60.4	66.7	60.4	52.4	41.8	30.6	23.9
	10%	DB	16.3	19.0	30.6	39.1	51.3	67.0	73.5	65.0	53.7	41.3	29.1	21.3
		MCWB	15.3	17.5	27.7	34.9	44.6	58.6	64.9	58.8	49.7	38.7	27.4	20.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	29.7	36.0	47.7	58.1	68.6	73.1	68.0	61.1	49.8	35.4	30.7
		MCDB	32.1	32.1	40.6	56.3	69.1	76.5	80.8	75.2	68.3	53.6	36.7	32.5
	2%	WB	23.4	25.6	33.0	40.1	53.5	64.5	70.4	64.3	56.4	45.0	33.2	27.3
		MCDB	24.8	27.2	36.3	45.2	63.8	72.8	78.1	70.3	61.2	48.2	34.6	28.9
	5%	WB	19.2	21.5	30.6	37.4	49.4	61.7	68.0	61.8	53.0	41.9	30.6	24.0
		MCDB	20.5	22.9	33.8	41.2	57.3	69.9	75.2	66.8	56.6	44.6	32.0	25.2
	10%	WB	15.3	17.5	27.7	35.2	44.9	59.0	65.8	59.4	50.3	39.0	27.3	20.2
		MCDB	16.3	19.0	30.5	38.9	50.9	66.1	72.5	64.0	53.3	41.0	29.0	21.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	14.8	16.8	15.6	13.7	13.8	14.9	14.1	11.6	9.4	12.9	14.3
		MCDBR	16.0	15.1	16.0	16.4	23.6	20.7	20.6	19.5	17.6	12.2	11.9	15.4
		MCWBR	15.0	13.7	12.8	11.6	14.4	11.4	11.1	12.0	11.6	9.2	10.4	14.2
	5% WB	MCDBR	15.4	15.1	15.7	14.7	22.6	18.4	18.2	17.3	16.2	12.0	11.2	14.9
		MCWBR	14.8	13.9	13.1	10.9	14.3	11.5	11.3	11.8	11.5	9.5	10.4	14.0
(40) Clear-Sky Solar Irradiance	taub		0.206	0.266	0.313	0.340	0.352	0.368	0.411	0.375	0.328	0.286	0.204	0.213
	taud		1.893	2.048	2.051	2.091	2.293	2.435	2.298	2.376	2.474	2.280	1.931	1.902
	Ebn at Noon		82	188	232	258	269	268	249	248	238	185	79	14
	Edh at Noon		7	19	31	38	35	31	35	29	21	15	6	2
(44) All-Sky Solar Radiation	RadAvg		33	228	724	1342	1703	1664	1671	1095	595	233	57	9
	RadStd		3	23	43	86	113	132	160	89	60	28	7	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	+2.09	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page